LESSON 9.1

Explore Relationships Between Two Variables

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
independent variable *Spanish: variable independiente*	The input quantity in a model, chosen or observed.	hours studied
dependent variable *Spanish: variable dependiente*	The quantity that changes in response to the independent variable.	quiz score
depends on *Spanish: depende de*	One quantity depends on another when changing the first causes the second to change.	Score depends on study time.
quantity *Spanish: cantidad*	Something that can be counted or measured.	minutes, dollars, laps
relationship *Spanish: relación*	How two quantities change together.	more time → higher score
Table of values *Spanish: Tabla de valores*	A list of matched input and output pairs.	x: 2, 4 paired with y: 14, 28

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bike rental costs $5 for each hour. Name the variables.

1. The quantities are hours rented and total cost.
2. Cost changes when hours change.
3. “Total cost depends on hours rented” makes sense.

Answer: hours = independent; cost = dependent

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Claire burned 123 Calories on a 30-minute jog and 246 Calories on a 60-minute jog.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The independent variable is _____ because you _____ it.
- The dependent variable is _____ because it _____ .
- _____ depends on _____ .
- If the _____ changes, then the _____ changes in response.

Word bank independent · dependent · variable · quantity · depends on · input · output · cause · respond · choose · changes · relationship

WATCH OUT!

A common mistake

Naming the variables backward

LESSON 9.1

Explore Relationships Between Two Variables

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

1 **Circle the letter of the best answer. Show how you know.**

Akela chooses how many hours of sunlight each tray gets. Which is the INDEPENDENT variable?

(A) Plant height
(B) Hours of sunlight
(C) Number of trays
(D) Number of seeds

HINT Which quantity did Akela get to decide?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

In Akela's experiment, which is the DEPENDENT variable?

(A) Hours of sunlight
(B) The kind of seed
(C) The size of the tray
(D) Plant height

HINT Which quantity RESPONDED to the other?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Claire burned 123 Calories on a 30-minute jog. Which quantity is subject to choice?

(A) Calories she burns
(B) Her heart rate
(C) Minutes she jogs
(D) The weather

HINT What does Claire decide before she starts?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which sentence is written in the correct order?

(A) Minutes jogged depends on Calories burned.
(B) Calories burned depends on Calories burned.
(C) Neither quantity depends on the other.
(D) Calories burned depends on minutes jogged.

HINT Say both versions out loud. Which one is true?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A tray that got 2 hours of sun grew to 4 cm. A tray that got 6 hours grew to 12 cm. As sunlight increases, plant height…

(A) decreases
(B) stays the same
(C) increases
(D) cannot be determined

HINT Compare the two trays.

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

You buy tickets at $45 each. Which quantity depends on the other?

(A) Total cost depends on number of tickets

(B) Number of tickets depends on total cost

(C) The price per ticket depends on the total cost

(D) Nothing depends on anything here

HINT Which number do you pick at the counter?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you and your partner put in different columns. Say the sentence 'The ____ depends on the ____' out loud both ways — which version is true, and how does that settle which variable is independent?

SENTENCE STARTER The ___ depends on the ___, so ___ is the independent variable.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 9.1

Explore Relationships Between Two Variables

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

1 **Circle the letter of the best answer. Show how you know.**

A gym charges $24.95 per month. Which pair correctly names the variables?

(A) Independent: months; Dependent: total cost
(B) Independent: total cost; Dependent: months
(C) Independent: $24.95; Dependent: months
(D) Independent: months; Dependent: $24.95

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Claire jogs 30 minutes and burns 123 Calories. At the same rate, how many Calories would a 90-minute jog burn?

(A) 246
(B) 300
(C) 369
(D) 410

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which of these pairs is NOT a dependent relationship?

(A) Number of pizzas sold and weekly cost of cheese

(B) Number of pizzas sold and number of employees the shop has

(C) Hours jogged and Calories burned

(D) Tickets bought and total cost

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A plant grows 2 cm for every hour of daily sunlight. Which table row is correct?

(A) 4 hours to 8 cm

(B) 4 hours to 6 cm

(C) 4 hours to 2 cm

(D) 4 hours to 4 cm

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which question would you ask to find the DEPENDENT variable?

(A) Which quantity responds when the other one changes?

(B) Which quantity is larger?

(C) Which quantity has units?

(D) Which quantity comes first alphabetically?

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Claire decides to walk instead of jog for the same 30 minutes. What happens to the two variables?

(A) Calories burned becomes the independent variable

(B) Neither is a variable any more

(C) Time stays the independent variable; Calories burned still depends on it

(D) Time becomes the dependent variable

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you and your partner put in different columns. Say the sentence 'The ____ depends on the ____' out loud both ways — which version is true, and how does that settle which variable is independent?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 9.1

Explore Relationships Between Two Variables

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Two quantities are related. Which statement is ALWAYS true?

(A) The independent variable is always the larger number

(B) The independent variable is the input used to describe the relationship

(C) The dependent variable is always measured in dollars

(D) The independent variable is always listed first in a table

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each scenario to its (independent → dependent) pair.

_____ Gym membership

_____ Seedling experiment

_____ Jogging

_____ Buying movie tickets

(A) months → total cost

(B) hours of sunlight → plant height

(C) minutes → Calories burned

(D) tickets → total price

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Give an example where the SAME quantity can be independent in one problem and dependent in another.

(A) Time is always independent

(B) Time: chosen when you pick how long to jog, but measured as a result when you run a set distance

(C) Time is always dependent

(D) A quantity can never switch roles

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The pizza shop spends $1.15 on cheese per pizza. Which weekly total is correct if it makes 200 pizzas?

(A) $115

(B) $201.15

(C) $230

(D) $1,150

PER 1

5 **Circle the letter of the best answer. Show how you know.**

Which relationship is NOT a case of one quantity depending on the other?

(A) Hours studied and quiz score

(B) Minutes jogged and Calories burned

(C) Pizzas made and cheese used

(D) A student's height and their house number

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Write the relationship as a sentence: a printer prints 12 pages per minute.

(A) Minutes running depends on pages printed

(B) 12 depends on the number of pages

(C) Pages printed depends on minutes running

(D) The printer depends on the pages

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you and your partner put in different columns. Say the sentence 'The ____ depends on the ____' out loud both ways — which version is true, and how does that settle which variable is independent?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help